Muhammad Muizz Zafar

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Security Clearance

Secret Level Security Clearance Jul 2026
Enhanced Reliability Level Security Clearance March 2024

Education

University of Ottawa Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* Sept 2020 - Dec 2024

Work experience

Software Analyst May 2024 - Present
Elections Canada Ottawa, ON

- Executed enterprise database migration by upgrading **SQL Server 2019 to SQL Server 2022** across **DEVTEST, UAT, and PROD** environments, migrating **50M+ records**, database schemas, stored procedures, triggers, and security configurations with zero downtime.
- Refactored legacy SQL code to modern **T-SQL standards**, updating deprecated syntax and optimizing database objects including stored procedures, views, and functions.
- Managed **role-based database access control** by configuring user permissions, roles, and security policies across multiple databases in accordance with government compliance requirements.
- Implemented **database backup, restore, and recovery procedures** across environments to support system reliability and disaster recovery readiness.
- Developed a reusable ETL validation framework using **SQL (T-SQL)**, **Python** and **Pandas**, enabling automated validation of large datasets (**20M+ records**) across dev, uat, and prod and created visual dashboards for data quality KPIs using **Streamlit**
- Built and maintained scalable **ETL pipelines** using **Python**, and **SQL** to extract data from **Oracle** databases, transform it, and load it into MSSQL data warehouse
- Built internal **data analytics dashboards** using **PowerBI** to surface database health, operational metrics, validation trends, and data quality issues for directorate and senior management.
- Assisted with **database monitoring and maintenance**, including performance diagnostics, log analysis, and troubleshooting database issues across enterprise systems.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of production data infrastructure across shared environments.

Full Stack Developer September 2023 - April 2024
University of Ottawa Ottawa, ON

- Built and maintained **JAVA** REST APIs and optimized database queries to improve response times for a platform serving **20,000+ users**.
- Implemented automated **unit and end-to-end tests** using Cypress, Jasmine, and JUnit to ensure application reliability.
- Investigated and resolved production issues by debugging backend APIs and database queries, improving system reliability.

Cloud Developer Intern May 2023 - August 2023
Trend Micro Inc. Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **Golang**, **AWS CDK**, through **CloudFormation** to provision **50+ AWS resources** and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput and strengthen fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across **50,000+ cloud artifacts per day**.

Cloud Developer Intern September 2022 - December 2022
Trend Micro Inc. Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.